DPP-INDUSTRIAL

A Tokenless Verifiable-Credential Architecture for the European Digital Product Passport in Industrial Measurement Equipment

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ABSTRACT

The European Ecodesign for Sustainable Products Regulation (ESPR) mandates a Digital Product Passport (DPP) for a growing set of product categories beginning 19 July 2026 with the launch of the EU Central DPP Registry. Existing DPP implementations are predominantly proprietary, narrowly vertical, or rely on Web3 architectures that conflict with the regulatory and reputational constraints of industrial B2B markets. This paper presents DPP-Industrial, a tokenless verifiable-credential architecture for ISO-17025-class measurement equipment that simultaneously satisfies ESPR, the EU Machinery Regulation 2023/1230, and the General Data Protection Regulation (GDPR). The architecture composes W3C Verifiable Credentials 2.0, the Ethereum Attestation Service on Base L2, ERC-5192 soulbound tokens, NXP NTAG 424 DNA cryptographic NFC tags, and Noir-based zero-knowledge proofs into a nine-layer system. We define six lifecycle attestation types aligned with the CIRPASS-2 Core Ontology, present three end-to-end protocol flows (asset onboarding, calibration event, and third-party verification), and derive a cost model showing an estimated 0.79 USD per asset across a ten-year lifecycle. We discuss open questions around secondary-market resale of soulbound assets, EU Central DPP Registry integration, and the legal status of cryptographic deletion under GDPR Article 17. The system is designed for incremental deployment through five minimum viable products (MVPs) across thirty months, with industrial-instrument calibration as the initial wedge use case.

Keywords: *Digital Product Passport; ESPR; Verifiable Credentials; Ethereum Attestation Service; Soulbound Tokens; Industrial IoT; GDPR; Selective Disclosure; Zero-Knowledge Proofs; Trust Anchor; ISO 17025; CIRPASS-2.*

1. Introduction

1.1 Regulatory Context

The European Union has, since the adoption of the European Green Deal in 2019, progressively expanded the scope of product-level transparency obligations on industrial goods placed on the EU market. The Ecodesign for Sustainable Products Regulation, EU 2024/1781 [1], introduces a horizontal mandate for a Digital Product Passport across all categories specified by delegated acts. Battery passports become mandatory on 18 February 2027 [1, 2]. Textiles, iron and steel, furniture, electronics, and construction products follow on staggered timelines through 2030. In parallel, the EU Machinery Regulation 2023/1230 [3] imposes a digital-documentation requirement on industrial machinery placed on the market after January 2027.

The Central DPP Registry, hosted by the European Commission, goes live on 19 July 2026 and provides a uniform identifier service and discovery layer for all DPPs across all in-scope verticals. Economic operators are required to register every in-scope product and to provide a Backup DPP Service compliant with ESPR Article 9. The Registry does not itself host product data; it acts as a directory [1, 4], which leaves the substantive design of the DPP itself to economic operators within the constraints of the Core Ontology being defined by the CIRPASS-2 consortium [5].

1.2 Problem Statement

Industrial measurement equipment occupies an unusual position in the DPP landscape. The category is not yet explicitly delegated under ESPR, but it intersects the Machinery Regulation digital-documentation requirements and, through the European harmonized standard EN 60751 and ISO 17025 accreditation requirements [6], maintains long-lived per-asset records of conformity, calibration, repair, and decommissioning that map naturally onto a DPP. Existing solutions for industrial-equipment lifecycle records are paper-bound, hosted in vendor-locked product-management systems (PMS), or in proprietary cloud silos with limited interoperability. The first industrial company that ships an EU-Registry-compatible DPP for measurement equipment will have a non-trivial advantage when the Machinery Regulation digital-documentation requirement becomes enforceable in 2027.

Public DPP implementations to date fall into three categories. First, enterprise-compliance systems such as iPoint and SAP Green Token offer DPP modules bolted onto ERP infrastructure [7], which work for the largest manufacturers but exhibit poor interoperability and lack credentialing. Second, vertically scoped startups such as Circularise (chemicals, plastics), Madaster (construction materials), and Fluxy.One (textiles) implement DPP-like functionality in their respective verticals [8] but typically rely on centralized data stores. Third, Web3-native projects such as Spherity and Minespider apply Self-Sovereign Identity (SSI) concepts to DPP problems [9], but their adoption among industrial buyers is constrained by token-related branding and operational complexity. None of these offerings combines (a) vertical depth in industrial measurement, (b) cryptographically anchored verifiable credentials, and (c) a tokenless, GDPR-aligned, B2B-suitable architecture.

1.3 Contributions

This paper makes the following contributions:

- **A complete architectural pattern** for a regulatory-compliant Digital Product Passport for industrial measurement equipment, composed from existing standards (W3C VC 2.0, EAS, ERC-5192, GS1 Digital Link, NTAG 424 DNA, Noir/Mopro).
- **A formal data model** consisting of six lifecycle attestation types aligned with the CIRPASS-2 Core Ontology, with explicit referential linkage producing a cryptographically verifiable asset history.
- **A cryptographic trust-anchor protocol** for physical-to-digital binding using NTAG 424 DNA Secure Unique NFC authentication, which solves the long-standing weak-link problem in physical-asset identification systems.
- **A privacy architecture** combining off-chain encrypted storage, SD-JWT selective disclosure, and Noir zero-knowledge proofs, addressing GDPR Article 17 compatibility through cryptographic deletion as documented in EDPB draft guidance on blockchain and personal data.
- **An empirical cost model** showing an estimated 0.79 USD per asset across a ten-year lifecycle, demonstrating commercial viability of the architecture at modest production volumes.
- **An implementation methodology** based on five sequential MVPs over thirty months, with explicit hypotheses, exit criteria, and validation partners.

The remainder of the paper is organized as follows. Section 2 reviews related work on DPP frameworks, verifiable credentials, and attestation systems. Section 3 introduces the design principles and overall system architecture. Section 4 specifies the data model. Section 5 details the physical-to-digital trust anchor. Section 6 describes the privacy and compliance architecture. Section 7 presents the three primary protocol flows. Sections 8 and 9 analyze cost and security respectively. Section 10 describes the incremental implementation methodology. Section 11 discusses limitations and open questions. Section 12 concludes.

2. Background and Related Work

2.1 The European Digital Product Passport Framework

The Digital Product Passport, as specified by ESPR [1], is a structured data set associated with a physical product that contains, at minimum, information on product identification, material composition, repairability, recyclability, environmental footprint, and end-of-life handling. The DPP is accessed via a unique product identifier rendered on or within the product, typically via a data carrier compliant with the GS1 Digital Link standard ISO/IEC 18975 [10]. The CIRPASS-2 consortium, funded under Horizon Europe, defines the Core Ontology and reference architecture, with deliverables released progressively from 2024 through 2026 [5]. The Core Ontology is, by design, extensible: vertical-specific schemas may extend the base vocabulary.

2.2 Verifiable Credentials and Decentralized Identifiers

The W3C Verifiable Credentials Data Model 2.0 [11], reaching W3C Recommendation status in 2025, defines a generic data model for cryptographically verifiable claims issued by an issuer to a holder

and verifiable by a third party. A Verifiable Credential is signed by the issuer, presented by the holder in a Verifiable Presentation, and verified against the issuer's public key. Decentralized Identifiers (DIDs) [12], a companion W3C Recommendation since 2022, provide a cryptographically verifiable identifier scheme that does not require a central registration authority. Multiple DID methods exist; in the architecture presented here, we use `did:web` for institutional issuers (resolved over HTTPS) and `did:pkh` for individual holders (derived deterministically from a public key).

2.3 Attestation Frameworks

The Ethereum Attestation Service (EAS) [13] is a tokenless on-chain protocol that provides the generic primitive: an entity attests to a claim about a subject. EAS attestations are recorded on Ethereum-compatible chains via a smart-contract registry, support a schema mechanism, allow revocation, and admit resolver contracts that can encode arbitrary business logic. EAS has been live on Base L2 [14] since 2023 and is used by Coinbase Verifications [15] and Gitcoin Passport for identity-related attestations. EAS differs from Verifiable Credentials in being chain-native (composable across attestations through refUID linkage) rather than file-based. In our architecture we combine both: EAS for on-chain composability and W3C VC 2.0 for regulator-facing document representation.

2.4 Soulbound Tokens

Buterin, Weyl, and Ohlhaver (2022) [16] proposed soulbound tokens as non-transferable tokens representing commitments, credentials, and affiliations rather than fungible value. EIP-5192 [17] standardizes the minimal interface: a token that is marked locked at mint, with no transfer function. Soulbound tokens have been widely deployed for credentialing and identity anchoring; in our architecture, ERC-5192 functions as a wallet-visible per-asset identity anchor that complements the off-wallet EAS attestations.

2.5 Prior DPP Implementations

Existing DPP-adjacent systems may be classified along three axes: vertical scope, data sovereignty, and verifiability. Table 1 summarizes the principal commercial and consortium-funded systems.

System	Vertical	Architecture	Decentralized?	Selective Disclosure?
Circularise	Chemicals, Plastics	Off-chain attestation	Partial	Yes
Spherity	General SSI	VC + DID, centralized issuer registry	Yes	Yes
Madaster	Construction materials	Centralized database	No	No
iPoint Suite	Enterprise compliance	Centralized SaaS	No	No
SAP Green Token	ERP-bound DPP	ERP-attached	No	Limited

System	Vertical	Architecture	Decentralized?	Selective Disclosure?
Minespider	Raw materials, Mining	Chain-anchored, upstream-only	Yes	Partial
Fluxy.One	Textiles	QR + centralized data	No	No
DPP-Industrial (this work)	Industrial measurement	VC + EAS + SBT, on-chain composable	Yes	Yes (default)

Table 1. *Comparison of existing DPP and DPP-adjacent systems with the architecture presented in this paper. Decentralization is assessed on the basis of data-store sovereignty (no single party can rewrite history). Selective disclosure refers to verifier interactions in which only the necessary claim subset is revealed.*

Spherity is the closest comparable in technical approach but operates as a horizontal identity platform rather than a vertically integrated DPP for industrial assets. Circularise and Madaster are vertically focused but rely on largely centralized data stores. SAP and iPoint offer enterprise-grade compliance but treat the DPP as a database export rather than an independently verifiable credential. The architecture presented in this paper occupies the previously unfilled position of a vertically focused, cryptographically anchored, tokenless DPP.

3. Design Principles and System Overview

3.1 Design Principles

The architecture is structured by five non-negotiable principles, each derived from a specific regulatory or operational constraint.

P1. Tokenless by design. At no point is a fungible or transferable token issued. The only on-chain token is ERC-5192 (soulbound), i.e. permanently non-transferable. The rationale is threefold: (i) the General Data Protection Regulation imposes complex joint-controller obligations on transferable on-chain tokens; (ii) the Markets in Crypto-Assets Regulation [18] grants a categorical carve-out for non-fungible soulbound tokens in Recital 11 but would otherwise treat transferable tokens as crypto-asset service providers requiring CASP licensing; (iii) industrial buyers (plant managers, compliance officers, accredited calibration labs) are reputationally averse to token communication after consumer Web3 failures.

P2. Off-chain payloads, on-chain hashes. Personal or commercially sensitive data never appears on chain. On chain there are only (i) hashes of off-chain stored payloads, (ii) schema hashes, (iii) issuer DIDs, (iv) expiration and revocation status. Rationale: the EDPB's draft guidance on personal data and blockchain technologies [19] holds that technical impossibility cannot be invoked to justify non-compliance with Article 17 GDPR. Cryptographic deletion (key destruction) functionally satisfies Article 17 even though the encrypted on-chain or persistent-storage blob remains; this practice is broadly tolerated though not yet judicially confirmed.

P3. Issuer = real-world authority. An attestation has only the trust value of its issuer. The platform is never the issuer of substantive claims; it is infrastructure. Manufacture attestations are signed by the manufacturer, Conformity attestations by notified bodies, Calibration attestations by DAkkS-

accredited laboratories [20]. Each issuer carries an established accreditation trust anchor and an existing business model that the architecture amplifies rather than replaces.

P4. EU Registry-compatible, not a substitute. The architecture complements the EU Central DPP Registry rather than replacing it. Every asset is registered in the Registry (a legal obligation) and additionally verifiably attested on chain (an added value). The on-chain attestation extends, rather than substitutes for, the official Registry record. This avoids any legal-substitution risk.

P5. Selective disclosure as default. Every verifier query is mediated by default through selective disclosure (SD-JWT [21]) or a zero-knowledge proof, not by transmission of the full credential. Full disclosure is explicitly opt-in by the holder.

3.2 Actor Model

The system has three principal actor classes, derived from the Verifiable Credentials trust triangle [11]:

- **Issuers** — institutional entities authorized to make substantive claims about assets. In our setting these comprise manufacturers, notified bodies (e.g. TÜV, DEKRA), DAkkS-accredited calibration laboratories, authorized service partners, and authorized recyclers. Each issuer is identified by a did:web DID resolvable over HTTPS.
- **Holders** — entities that hold custody of an asset and the associated credentials. In our setting these are operators of industrial measurement equipment, identified by a did:pkh DID derived from a smart-wallet address.
- **Verifiers** — entities that consume attestations to make trust decisions: insurers (asset-value underwriting), market-surveillance authorities (regulatory enforcement), auditors (ISO 17025 compliance), B2B-resale buyers, and service partners. Verifiers are identified at request time and need not pre-register with the system.

3.3 Layered Architecture

The system is organized into nine layers, shown in Figure 2 below. Seven of these layers are general-purpose verifiable-credential infrastructure that could in principle be reused for other product categories; the two DPP-specific layers (EU Registry Sync and Data Carrier) implement the regulatory adapter.

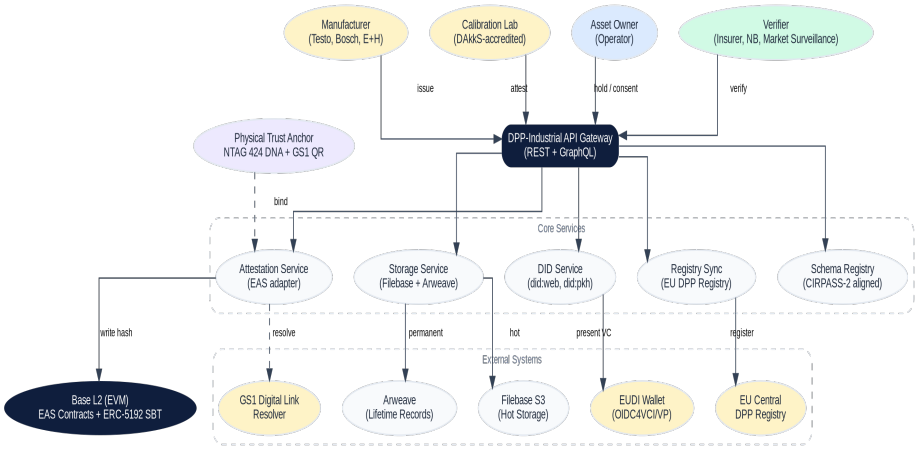

Figure 1. *System overview. Issuer, holder, and verifier actors interact through the API gateway with five core services. The physical trust anchor (NFC chip plus GS1 QR) and the EAS smart contracts on Base L2 anchor the on-chain identity. External integrations are with the EU Central DPP Registry, the EUDI Wallet, the GS1 Resolver network, and dual storage backends (Filebase and Arweave).*

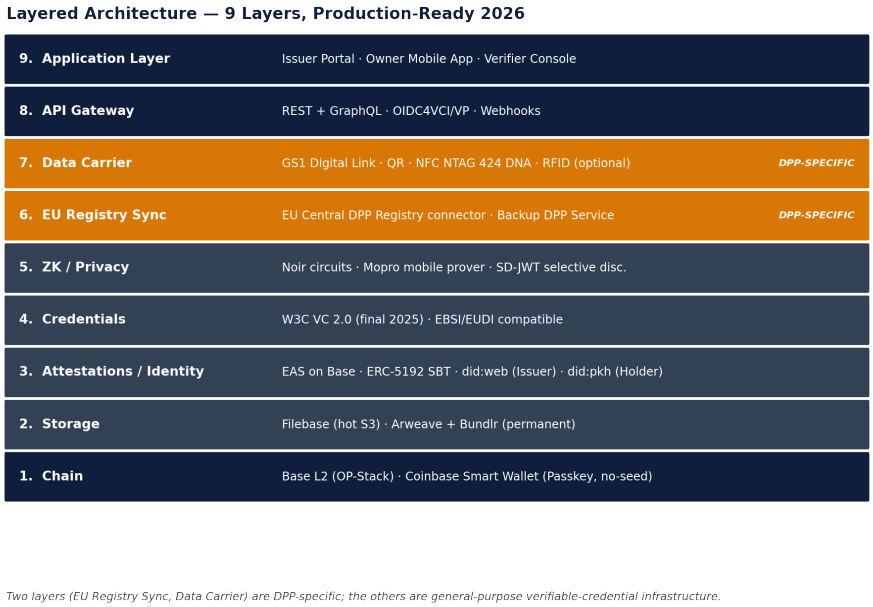

Figure 2. *Layered architecture. Layers 1–5 and 8–9 are general-purpose verifiable-credential infrastructure; layers 6 (EU Registry Sync) and 7 (Data Carrier) are the DPP-specific adapter.*

4. Data Model

4.1 Asset and Attestation Types

The data model is asset-centric: a single root Asset entity is referenced by six attestation types covering the full product lifecycle from manufacture to decommissioning. The model is shown in Figure 3. The Asset entity carries the immutable identifying attributes (DID, SBT token identifier, GTIN, serial number, NFC chip UID, manufacturer DID, creation timestamp); the six attestation types attach mutable lifecycle events through the EAS `refUID` mechanism [13].

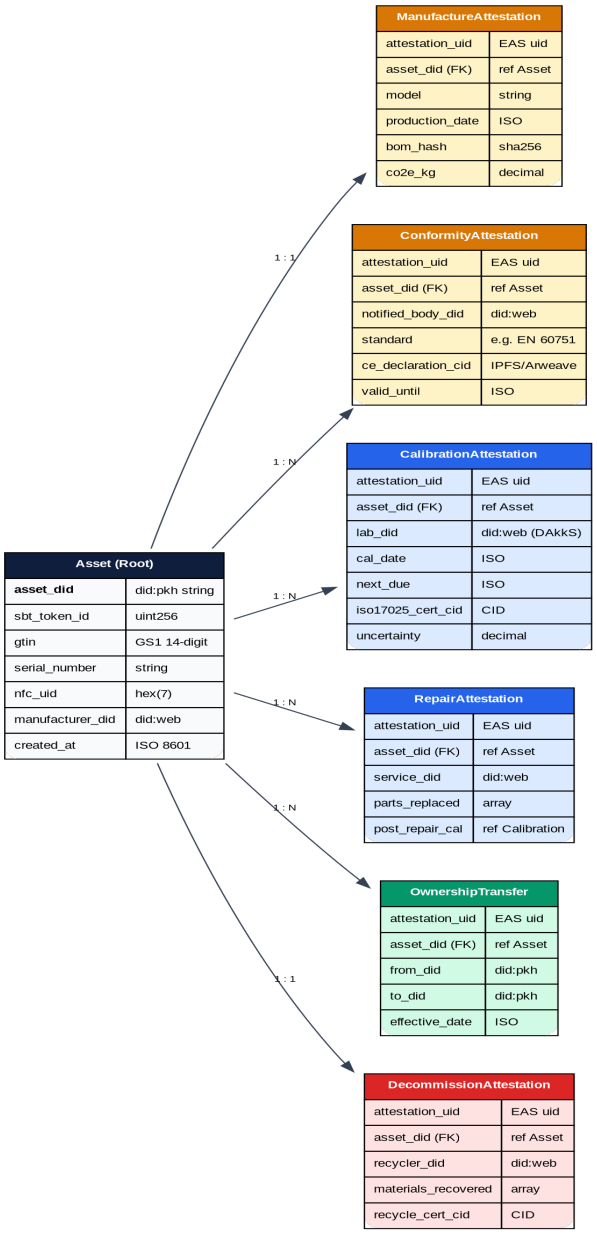

Figure 3. *Entity-relationship diagram of the data model. The Asset root is referenced by six attestation types, each linked through the EAS refUID mechanism to its predecessor in the asset lifecycle. ConformityAttestation, CalibrationAttestation, RepairAttestation, and OwnershipTransfer may occur multiple times per asset; ManufactureAttestation and DecommissionAttestation occur exactly once.*

The six attestation types and their semantic roles are:

- **ManufactureAttestation** — issued exactly once at production. Records the bill-of-materials hash, production date, model identifier, and a measured or computed CO_2 equivalent footprint.
- **ConformityAttestation** — issued by a notified body. Records the standard against which conformity was assessed (e.g. EN 60751 for resistance thermometers), the CE Declaration of Conformity payload hash, and the validity period.

- **CalibrationAttestation** — issued by a DAkkS-accredited calibration laboratory. Records the calibration date, the next due date, the ISO 17025 certificate identifier, and the measurement uncertainty achieved. This is the most frequent attestation type in the industrial context.
- **RepairAttestation** — issued by an authorized service partner. Records parts replaced and references any post-repair recalibration attestation.
- **OwnershipTransfer** — issued jointly by the outgoing and incoming owners. Records the effective transfer date and a reference to the previous OwnershipTransfer.
- **DecommissionAttestation** — issued exactly once by an accredited recycler. Records recovered materials, a reference to the recycling certificate, and triggers the resolver-contract retirement flag on the SBT.

4.2 Schema Versioning

EAS schemas are immutable after registration. Schema evolution is therefore handled by schema forking: a version-2 schema is registered as a new schema, and the original version-1 entry remains valid for historical attestations. Verifier code maps the older schema UID to a legacy adapter; new attestations are written against the latest schema. This pattern decouples data evolution from smart-contract upgrades, which by design remain rare and explicitly governed by the MultiSig [13].

4.3 Example: CalibrationAttestation

The CalibrationAttestation is reproduced here in full, both as the on-chain EAS schema definition and as the off-chain W3C VC 2.0 payload [11]:

```
// EAS Schema (on-chain, immutable after registration)
schema CalibrationAttestation {
    bytes32 assetDid;            // refers to root Asset SBT
    bytes32 labDid;              // did:web of accredited lab
    uint64  calDate;             // Unix timestamp
    uint64  nextDueDate;
    bytes32 isoCertCid;          // off-chain ISO 17025 cert CID
    bytes32 payloadCid;          // full calibration report (encrypted)
    bytes32 payloadHash;         // sha256 of unencrypted report
    uint32  uncertaintyMicro;    // measurement uncertainty * 1e6
    uint8   status;              // 0 = valid, 1 = revoked, 2 = superseded
}

// Off-chain payload (W3C VC 2.0, encrypted at rest)
{
  "@context": ["https://www.w3.org/ns/credentials/v2"],
  "type": ["VerifiableCredential", "CalibrationCredential"],
  "issuer": "did:web:dakks-lab-12345.de",
  "validFrom": "2026-05-15T08:30:00Z",
  "validUntil": "2027-05-15T08:30:00Z",
  "credentialSubject": {
    "id": "did:pkh:eip155:8453:0xabc...",
    "assetType": "Thermometer testo 175 H1",
    "standard": "EN 60751",
    "measurementPoints": [
      { "nominal": -20.0, "actual": -19.87, "uncertainty": 0.12 },
      { "nominal":   0.0, "actual":  -0.03, "uncertainty": 0.08 },
      { "nominal":  50.0, "actual":  50.04, "uncertainty": 0.09 },
      { "nominal": 100.0, "actual": 100.11, "uncertainty": 0.11 }
    ],
```

```
    "ambientConditions": { "temp": 23.1, "humidity": 47 },
    "labAccreditation": "D-K-15012-01-00"
  },
  "proof": {
    "type": "DataIntegrityProof",
    "cryptosuite": "eddsa-rdfc-2022",
    "verificationMethod": "did:web:dakks-lab-12345.de#key-1",
    "proofPurpose": "assertionMethod",
    "proofValue": "z58..."
  }
}
```

The on-chain schema is deliberately minimal: it contains only what is needed to verify the validity, integrity, and authenticity of the off-chain payload. The off-chain payload contains the full calibration record in a regulator-recognizable form, signed by the accredited laboratory's did:web key using the EdDSA cryptographic suite of the W3C Verifiable Credentials Data Integrity specification [11].

5. The Physical-to-Digital Trust Anchor

5.1 The Binding Problem

A Digital Product Passport that cannot be reliably bound to its physical asset is of little value: an attacker who clones the carrier can clone the credential. In existing industrial-asset identification systems (notably ISO 11784/85 RFID transponders used in livestock and inventory contexts), the carrier is a passive tag with a readable but unprotected serial number, clonable in under a minute with commodity hardware. ESPR Article 11 [1] requires that the data carrier be tamper-resistant and uniquely associated with the product, but does not specify a cryptographic standard. The architecture presented here uses a hardware-anchored cryptographic primitive that elevates the binding to the level of computational security.

5.2 NTAG 424 DNA and Secure Unique NFC Authentication

We adopt the NXP NTAG 424 DNA [22] as the canonical NFC data carrier. The NTAG 424 DNA is a consumer-priced NFC chip (USD 0.30–1.50 per unit depending on volume and encapsulation) that embeds an AES-128 hardware key in a tamper-resistant secure element. The chip implements Secure Dynamic Messaging (SDM), in which each tap of the chip produces a Secure Unique NFC (SUN) URL parameter consisting of (i) the chip's static UID, (ii) a monotonic counter, and (iii) an AES-128 cryptographic MAC over these values keyed by the in-chip secret. The verifier (our backend) re-derives the expected MAC using a paired key and rejects taps whose counter does not strictly increase, defeating replay.

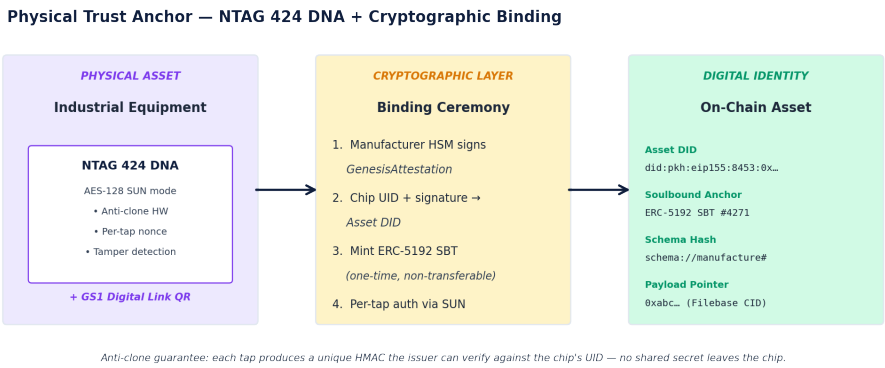

Figure 4. *The physical-to-digital trust anchor. A binding ceremony at production links the NTAG 424 DNA chip's hardware-protected secret to a manufacturer-signed GenesisAttestation, an ERC-5192 soulbound token, and an off-chain payload. Each subsequent tap of the chip produces a cryptographically unique authentication code that the backend verifies.*

5.3 Binding Ceremony

At asset production, a binding ceremony links the physical chip to the digital identity in four steps:

- **Step 1.** The manufacturer's HSM-resident did:web private key signs a GenesisAttestation containing the chip UID, the bill-of-materials hash, the production timestamp, and the model identifier.
- **Step 2.** An ERC-5192 soulbound token is minted on Base L2 with token-ID = chip UID, owner = manufacturer wallet, and metadata pointing to the (encrypted) genesis payload CID.
- **Step 3.** The EAS attestation is written on Base with refUID = SBT token-ID and the on-chain schema set to ManufactureAttestation.
- **Step 4.** The asset is registered in the EU Central DPP Registry with manufacturer-DID, GTIN, serial number, and the URI to the DPP service.

After the ceremony, the binding between the physical chip and the digital identity is reduced to the hardware secret embedded in the chip's secure element. An attacker would need to extract the AES-128 key from the secure element, which under current public knowledge requires destructive physical analysis with success uncertain even with state-of-the-art side-channel laboratories [22]. Cloning by simple electronic reading is not possible: a clone chip would have the correct UID but the wrong secret key, and would produce an invalid SUN MAC on the first tap.

6. Privacy and Compliance Architecture

6.1 Off-Chain Payloads with On-Chain Hashes

The architecture treats the blockchain layer as an immutable notary, not as a database. The substantive content of each attestation is stored off-chain in two parallel backends: Filebase (S3-API IPFS provider, geo-redundant, hot-access) [23] and Arweave with the Bundlr settlement layer (pay-once, multi-decade persistence) [24]. The on-chain attestation contains only the payload hash, the schema UID, the issuer DID, the validity period, and the revocation status. This pattern restricts on-chain data to information that has no plausible interpretation as personal data under GDPR Article 4

[25]. Encryption of off-chain payloads uses AES-256-GCM with per-asset symmetric keys managed in the manufacturer HSM.

6.2 Selective Disclosure via SD-JWT

Selective Disclosure JWT [21] enables the holder to present an arbitrary subset of credential claims to a verifier. In our architecture, the canonical use case is an insurer requesting confirmation that a calibration is currently valid: the holder can produce an SD-JWT-VP disclosing only the `validUntil` field and a derivative measurement-uncertainty bound, without revealing the full calibration report. The holder's mobile application generates the SD-JWT-VP locally, signing with the wallet key (anchored in a smartphone Passkey via WebAuthn). Verifier-side validation requires only the issuer's public key, fetched from the issuer's did:web.

6.3 Zero-Knowledge Proofs

For predicates that cannot be expressed as a simple subset of credential fields, the architecture supports zero-knowledge proofs over the credential data. We adopt Noir [26] as the circuit language: Noir is a Rust-like domain-specific language compiled to a PLONK-style backend [27] with a universal trusted setup, eliminating the need for per-circuit trusted-setup ceremonies that would otherwise impose substantial audit cost. Proof generation on the holder's smartphone uses Mopro [28], a mobile prover runtime that achieves 2–8 second proof generation for typical credential predicates on modern smartphones.

Three production zero-knowledge predicates are envisaged in the initial deployment:

- **CalibrationValidityProof** — proves that the most recent CalibrationAttestation for the asset is not older than N months and that the measurement uncertainty is below a verifier-specified bound, without revealing the date or the uncertainty value.
- **OwnershipProof** — proves that the prover is the current owner of an asset belonging to a verifier-specified registered fleet, without revealing owner identity or asset identity.
- **AccreditationProof** — proves that an attestation was issued by an accredited body of a specified type (e.g. “any DAkkS-accredited laboratory”) without revealing which one.

6.4 GDPR Compatibility

The architecture is designed to satisfy GDPR through three complementary mechanisms. First, by P2 (off-chain payloads), personal data never appears on chain and therefore cannot trigger the on-chain immutability problem. Second, per-personal-data encryption keys are managed in HSMs and may be deterministically destroyed; the EDPB's draft guidance on personal data in blockchain technologies acknowledges cryptographic deletion as a functional equivalent of erasure under Article 17, although this position is not yet judicially confirmed [19]. Third, joint-controllership relationships among the manufacturer (data controller), platform operator (data processor or joint controller depending on legal characterization), and the EU Central DPP Registry are governed by an Article 26 joint-controller agreement signed at issuer onboarding.

7. Protocol Flows

This section presents three end-to-end protocol flows that exercise the principal interactions of the system: asset onboarding (issuer-initiated), calibration event (issuer-initiated with physical asset verification), and verification (verifier-initiated).

7.1 Asset Onboarding

Asset onboarding is performed at the manufacturer (or OEM converter) at production time. A production batch of 500 devices can be fully onboarded in under ten minutes through the bulk-onboarding endpoint. The sequence is shown in Figure 5.

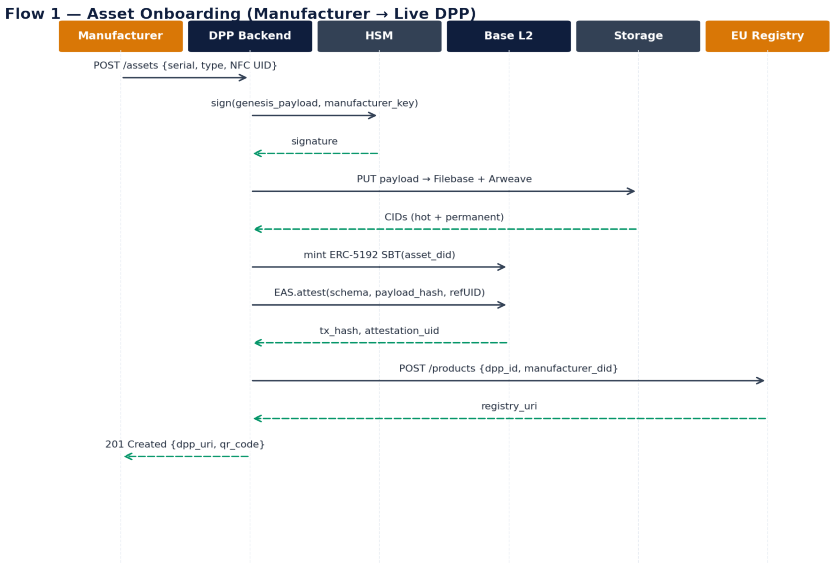

Figure 5. *Asset onboarding sequence. The manufacturer system invokes the bulk endpoint, the HSM produces the genesis signature, payloads are dual-written to Filebase and Arweave, the ERC-5192 SBT is minted and the EAS attestation is written on Base L2, and finally the asset is registered with the EU Central DPP Registry. The resulting QR code and NFC programming data are returned to the manufacturer.*

7.2 Calibration Event

Calibration events are the most frequent lifecycle interaction: industrial measurement equipment is calibrated one to four times per year per asset, depending on industry and standard. The protocol minimizes friction by integrating directly into the laboratory's existing PMS workflow rather than requiring a parallel interface. The sequence is shown in Figure 6.

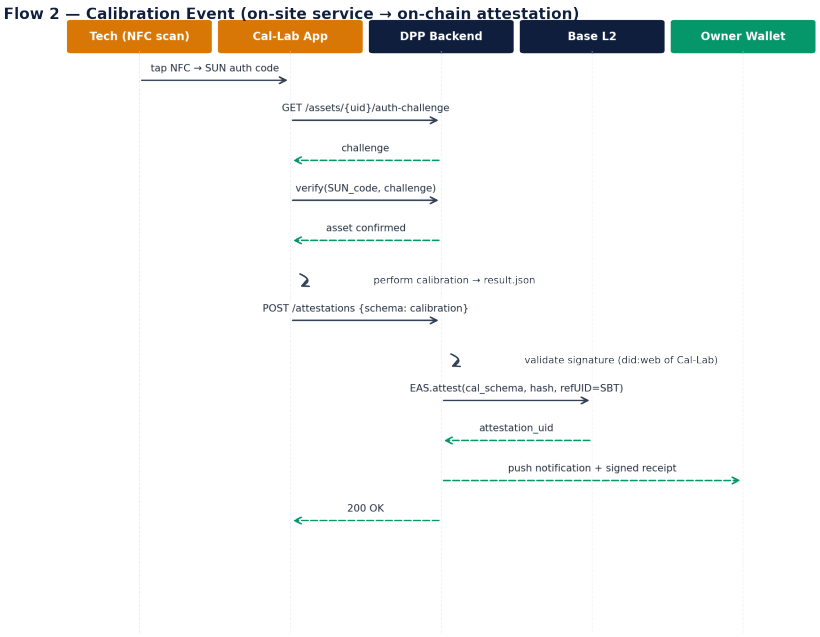

Figure 6. *Calibration event sequence. The technician taps the NFC tag, producing a SUN authentication code; the laboratory application verifies asset authenticity through the backend; the calibration is performed and recorded in a signed payload; the backend writes the attestation on Base L2 and notifies the owner wallet.*

The SUN authentication step is the key anti-spoofing mechanism. Without it, a calibration could in principle be recorded against an arbitrary asset DID; with it, the backend can confirm that the calibration was indeed performed at the physical asset to which the chip is bound.

7.3 Verification

Verification is initiated by a third party such as an insurer, market-surveillance inspector, B2B-resale buyer, or auditor. The verifier scans the QR code or taps the NFC tag with a standard smartphone running the public verifier application, which requires no account. The sequence is shown in Figure 7.

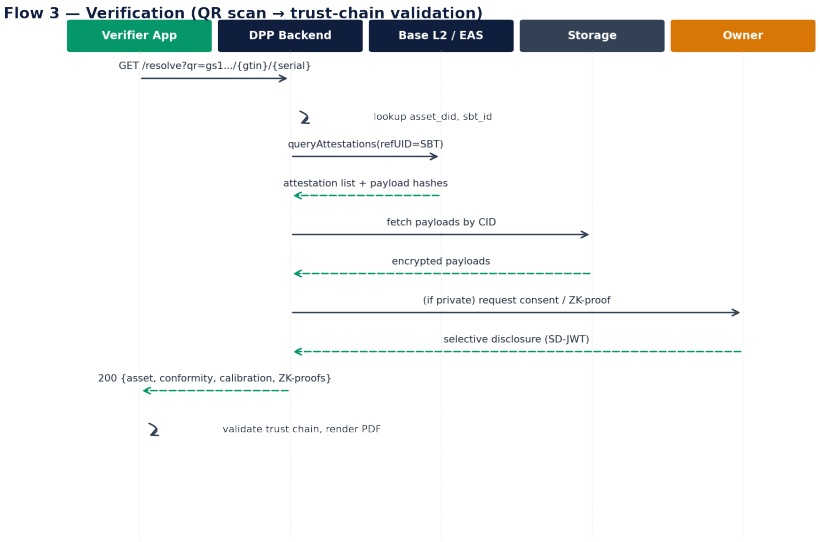

Figure 7. *Verification sequence. The verifier resolves the GS1 Digital Link URL; the backend looks up the asset DID and queries EAS for attestations referenced by the SBT; payloads are fetched from the storage backends; for sensitive disclosures, the holder's consent is solicited via the wallet; the verifier application validates the trust chain and renders a verification report.*

Two verification modes are supported. In the **public-resolution** mode, only non-sensitive metadata (manufacturer, model, conformity standard, and a Boolean validity flag for the most recent calibration) is returned. In the **selective-disclosure** mode, the backend forwards a consent request to the holder's wallet; the holder approves or denies within seconds, and the verifier receives an SD-JWT-VP or a ZK proof conveying only the requested predicate.

8. Cost and Performance Analysis

We model the cost of operating the system across a representative ten-year asset lifecycle. The model assumes twelve total attestations per asset (one ManufactureAttestation, one ConformityAttestation, eight CalibrationAttestations at annual cadence, two RepairAttestations or OwnershipTransfers, and one DecommissionAttestation), a per-attestation off-chain payload size of approximately 5 MB, and gas pricing characteristic of Base L2 in 2026 [14]. Table 2 summarizes the cost components.

Cost component	Unit cost	Per asset	Per 50k assets
Base L2 gas (EAS attest)	0.02 USD/attest	0.24 USD	12,000 USD
Base L2 gas (SBT mint)	0.05 USD/asset	0.05 USD	2,500 USD
Filebase hot storage	0.003 USD/MB·yr	0.15 USD	7,500 USD
Arweave permanent storage	0.007 USD/MB	0.35 USD	17,500 USD
NTAG 424 DNA + GS1 tag (hardware)	0.80 USD/asset	0.80 USD	40,000 USD (one-time)
Total software costs		**0.79 USD**	**~39,500 USD/yr**

Table 2. *Estimated cost components per asset across a representative ten-year lifecycle. Software cost (excluding one-time hardware) is approximately 0.79 USD per asset; at a deployment scale of 50,000 assets per year, recurring infrastructure cost is below 40,000 USD per year.*

Two observations: first, the total software cost per asset is small in absolute terms, and is dominated by Arweave permanent storage rather than on-chain gas. Second, the per-asset infrastructure cost is one to two orders of magnitude below typical European engineering and compliance budgets for the same asset category, confirming that the architecture is commercially viable at modest production volumes.

ZK proof generation. The Mopro mobile prover [28] benchmarks indicate that the three production circuits (CalibrationValidityProof, OwnershipProof, AccreditationProof) generate within 2–8 seconds on modern smartphones (iPhone 14 and Pixel 7 reference devices), with a memory footprint under 200 MB. Verification times are below 50 milliseconds. These latencies are compatible with interactive verification workflows in inspection contexts.

9. Security Analysis

9.1 Threat Model

We adopt a Dolev-Yao adversarial model for protocol-level threats and assume that the adversary has access to commercial NFC reading hardware but not state-funded side-channel laboratories. Table 3 enumerates the principal threats and the mitigations integrated into the architecture.

Threat	Attack vector	Mitigation
Tag cloning	Physical removal of NFC tag and electronic key extraction	NTAG 424 DNA AES-128 hardware secure element; tamper-evident encapsulation; EU Registry cross-check
Issuer compromise	Compromised laboratory or manufacturer signing key	Accreditation whitelist enforced by EAS resolver; MultiSig updates; notified-body audit trail; key rotation policy
Wallet key loss	Smartphone loss without Passkey cloud backup	Passkey cloud sync; social recovery (3-of-5 trusted contacts); manufacturer-mediated OwnershipTransfer recovery
Replay of SUN	SUN code captured by tag-cloning device, replayed	Backend counter check, sliding-window replay detection (1 hour)
Personal data leakage on-chain	Bug in attestation pipeline writes PII on-chain	Schema-validation pre-write gate; periodic data-classification audits; on-chain content is restricted to hashes and DIDs by design
Registry desynchronization	EU Registry status diverges from on-chain attestations	Backup DPP Service via independent provider; periodic sync reconciliation; reconciliation alarms

Table 3. *Threat model and mitigations. The Dolev-Yao model is adopted for protocol-level threats; physical threats requiring destructive side-channel analysis are assumed to be infeasible for the typical adversary.*

9.2 Cryptographic Audit Surface

Three smart contracts require external security audit prior to mainnet deployment: (i) the ERC-5192 SBT contract (used in its audited reference implementation form [17]); (ii) the EAS resolver contracts that encode the business rules (only accredited laboratories may write CalibrationAttestations, only the SBT holder may initiate ownership transfer); (iii) the issuer-registry multisignature contract. Estimated audit budget: 50–100 k€ at 2026 market rates from established Web3 security firms such as Trail of Bits, OpenZeppelin, or ChainSecurity.

10. Implementation Methodology

10.1 MVP-Based Incremental Deployment

The system is structured for incremental deployment through five sequential minimum viable products (MVPs) across thirty months. Each MVP isolates a distinct hypothesis and a single primary risk; success criteria are defined ex ante in measurable terms. The roadmap is shown in Figure 8.

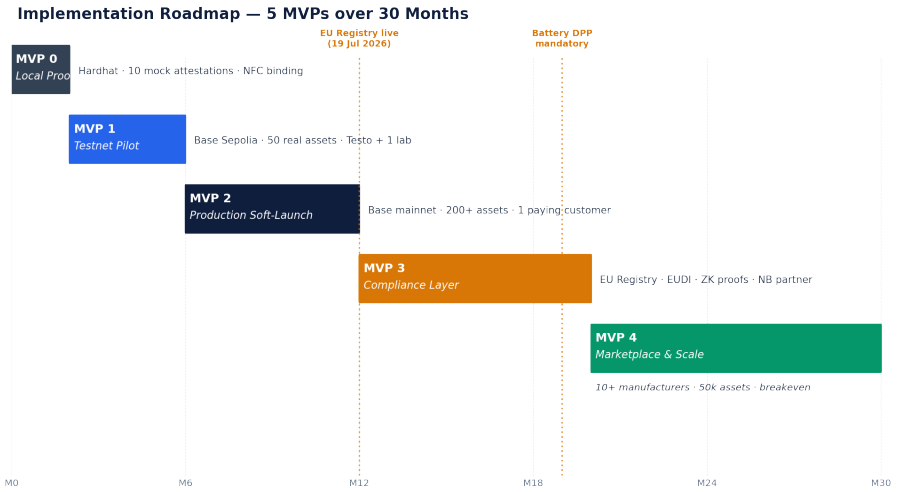

Figure 8. *Implementation roadmap with regulatory milestones. The five MVPs progress from local proof-of-concept through testnet pilot, production soft-launch, regulatory-compliance integration, and finally marketplace scale. The two amber-colored vertical lines indicate the EU Central DPP Registry go-live (19 July 2026) and the Battery Pass mandatory date (18 February 2027).*

MVP 0 — Local Proof (months 0–2). Hardhat-based EAS schema deployment, ERC-5192 mint, ten mock attestations, manually programmed NTAG tags, locally hosted REST API. Hypothesis: the stack functions end-to-end. Exit criterion: demonstrable working prototype convinces an internal manufacturer stakeholder to enter formal pilot discussions.

MVP 1 — Testnet Pilot (months 2–6). Base Sepolia deployment, Issuer Portal MVP, Owner Mobile App stub, Verifier web page with QR resolution. Pilot partners: one manufacturer (Testo as cornerstone) plus one DAkkS-accredited calibration laboratory. Asset volume: 50 real assets from a selected manufacturer product line. Hypothesis: a real laboratory can integrate the workflow without operator-level intervention. Exit criterion: laboratory uses the system unaided; one industrial insurer LOI signed.

MVP 2 — Production Soft-Launch (months 6–12). Base mainnet deployment, external security audit complete, full mobile applications for iOS and Android, Filebase plus Arweave dual write live, DPIA filed and accepted by the relevant Landesdatenschutzbehörde. Scaling: 200+ real assets, two manufacturers, three calibration laboratories. Hypothesis: the system handles production load and the first paying customer wires the first invoice. Exit criterion: first 50 k€ ARR with burn rate below 120 k€ per month.

MVP 3 — Compliance Layer (months 12–20). EU Central DPP Registry integration, Backup DPP Service contract executed, OpenID4VCI and OpenID4VP endpoints for EUDI Wallet [29, 30], ZK proof flows in production with CalibrationValidityProof as the first live use case. Notified-body partnership executed with TÜV or DEKRA. Scaling: five to ten manufacturers, 1,000+ assets, first EU geographic expansion (France, Italy, Netherlands). Hypothesis: the system is fully ESPR-compliant and integrates with the official EU identity infrastructure. Exit criterion: 10+ manufacturers in pipeline, ESPR compliance documented, Seed round prepared.

MVP 4 — Marketplace and Scale (months 20–30). Calibration-service marketplace integrated into the owner mobile application; insurer integration with a predictive-maintenance data feed; multi-vertical schema extension to laboratory and medical-device equipment. Scaling: ten or more manufacturers in production, 50,000+ assets, EU-wide geographic coverage. Hypothesis: the system is not only a compliance tool but a network-effect-enabled service marketplace with positive unit economics. Exit criterion: ARR run rate 3–8 M€, gross margin above 75 percent, logo retention above 95 percent.

10.2 Pilot Evaluation Design

The MVP 1 testnet pilot is designed to validate three measurable dimensions of system performance: (i) operator-friction reduction, measured as the additional time required per calibration event compared with the baseline PMS workflow; (ii) verification latency, measured as end-to-end time from QR scan to verification report rendering, including any consent-flow delay; (iii) error rate, measured as the proportion of failed onboarding or verification attempts attributable to system rather than user error. Acceptable thresholds for go/no-go to MVP 2 are operator-friction increment below 60 seconds per event, median verification latency below 3 seconds for public mode and below 12 seconds for selective-disclosure mode, and system-attributable error rate below 1 percent.

11. Discussion

11.1 Limitations

Soulbound non-transferability and the secondary market. Industrial assets are routinely traded on the B2B secondary market (equipment relocation, insolvency liquidation, second-life resale). The ERC-5192 SBT is non-transferable by construction. Ownership change is handled by the OwnershipTransfer attestation rather than token transfer, but this requires custodial cooperation that may not be available in all transfer scenarios (e.g., asset auction where the previous owner declines to co-sign). A wrapped-ownership pattern in which a separate transferable OwnershipNFT references the SBT is under design but complicates the trust chain and introduces custodian dependencies that themselves require auditing.

EU Central Registry API stability. The CIRPASS-2 consortium's final API specification is, as of the time of writing, in active consultation. Late-stage schema changes could invalidate parts of the RegistrySync layer. Mitigation is limited to active participation in the consortium working group and conservative schema modeling that anticipates likely revisions.

Backup DPP Service requirement. ESPR Article 9 requires that economic operators provide a Backup DPP Service operated independently of the primary service. A satisfactory commercial-or-cooperative partner has not yet been identified; candidates include national consortia (Bayern Innovativ, VDMA) or multi-provider arrangements analogous to the EUDI Wallet architecture.

Cryptographic deletion under GDPR Article 17. The EDPB's draft guidance on personal data and blockchain treats cryptographic deletion as a functional equivalent of erasure [19]. This position is not yet judicially confirmed, and a national supervisory authority could in principle take a stricter line, requiring physical deletion that is incompatible with Arweave-style permanent storage. Mitigation is limited to per-data-subject encryption and rapid key destruction on Article 17 request.

11.2 Open Technical Questions

Seven specific technical questions remain open and are flagged here for future work:

- Wrapped-ownership pattern design and audit (Section 11.1).
- CIRPASS-2 final API schema and migration plan from the working-draft schema to the final standard.
- Backup DPP Service partner identification and SLA negotiation.
- ZK circuit audit budget reduction through reuse of existing Aztec and Noir reference circuits.
- Smart-wallet vendor abstraction to mitigate Coinbase Smart Wallet lock-in risk.
- GS1 license cost modeling at scale, particularly for manufacturers that do not already hold a GS1 Company Prefix.
- Single-chain vs. multi-chain operational trade-off, including disaster-recovery strategy in the event of Base discontinuation.

12. Conclusion

This paper has presented DPP-Industrial, an architectural pattern for a Digital Product Passport for industrial measurement equipment that satisfies the European Ecodesign for Sustainable Products Regulation, the EU Machinery Regulation 2023/1230, and the General Data Protection Regulation. The architecture combines W3C Verifiable Credentials 2.0, the Ethereum Attestation Service on Base L2, ERC-5192 soulbound tokens, NXP NTAG 424 DNA cryptographic NFC tags, and Noir-based zero-knowledge proofs into a nine-layer system whose two regulatory-adapter layers are the only DPP-specific components.

The principal architectural contribution is the explicit composition of a cryptographic hardware trust anchor with a chain-native attestation framework, mediated by a soulbound identity token and complemented by selective-disclosure and zero-knowledge privacy mechanisms. The design satisfies five constraints simultaneously: regulatory compliance, GDPR alignment, tokenless branding,

complete verifiability, and incremental implementability. The cost model demonstrates a per-asset software cost of approximately 0.79 USD across a ten-year lifecycle, confirming commercial viability at deployment volumes consistent with mid-sized European industrial manufacturers.

Open questions remain in three areas: secondary-market resale of soulbound assets, finalization of the CIRPASS-2 Registry schema, and the legal status of cryptographic deletion under GDPR Article 17. Each of these is the subject of active design or regulatory consultation, and definitive resolution is expected within the timeframe of the proposed MVP 3 milestone in 2027.

Future work will extend the architecture in three directions: (i) generalization beyond measurement equipment to adjacent industrial verticals (laboratory and medical equipment, calibration-intensive food-safety equipment), (ii) empirical validation through the planned testnet pilot, and (iii) formalization of the selective-disclosure and zero-knowledge predicates against the emerging CIRPASS-2 Core Ontology.

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